

ClauseLens — RAG for Contract Intelligence

Gargi Sethi · Track D — RAG / LLM Knowledge Systems · Scenario 2 (Gen AI for Enterprise Documents) · github.com/gargisethi100/Avathon

1 · Problem & Domain

Legal teams drown in contracts and need natural-language answers about a *specific* document — “what is the termination-for-convenience notice period?” — with **citable**, **verifiable** support. This is a **knowledge-retrieval** problem, not skill acquisition: the knowledge (clause text) differs per document and grows over time. Hence **RAG (Track D)**, **not fine-tuning** (bakes stale knowledge into weights, cannot emit verbatim citations, needs retraining per contract) and **not agents** (orchestration cost, no retrieval-quality payoff for single-document lookup). **Corpus: CUAD** (510 commercial contracts, 41 clause categories, expert-annotated answer spans, CC BY 4.0). The spans give *real* retrieval ground truth, so quality is **measured**, not self-judged. Test partition: 102 contracts, 4,182 QA pairs — 70% unanswerable, treated as a large, honest abstention test.

2 · Approach & Algorithm Decisions (each decided by measured ablation; several priors refuted)

- **Chunking (Q15)** → **recursive-512, 15% overlap**. 512 beat 256 decisively (R@10 0.75 vs 0.67 — long clauses fragment at 256). Against our prior, a structure-aware clause-heading splitter did **not** beat recursive (near-tie), so we ship the simpler one. Boundary loss mitigated by overlap; verified **100% gold-span coverage** at every size — no answer is lost at a chunk boundary.
- **Embeddings (Q16)** → **bge-small (local, free)**. Empirically tied with 3×-larger bge-base (within ~1 pt R@10) → a free local model suffices. OpenAI rejected (second vendor); Bedrock-Cohere kept as optional cross-check.
- **Retrieval + fusion (Q17)** → **hybrid dense+BM25 with Reciprocal Rank Fusion**. **BM25 beat dense** (R@5 0.54 vs 0.47 — legal queries are keyword-heavy); hybrid-RRF edged both. A **MS-MARCO cross-encoder rerank hurt** (0.55→0.48): web-trained rerankers mis-transfer to contract language, so we ship without it (legal-domain reranker = future work). RRF over score-blending: rank fusion needs no normalization across incomparable cosine/BM25 scales. **FAISS flat** (exact; ANN needless at thousands of vectors; pgvector/HNSW is the production path).
- **Generation** → **Claude Haiku 4.5 on AWS Bedrock**. Grounding prompt: answer only from excerpts, quote verbatim, cite [n], else NOT_FOUND.
- **Hallucination defense (Q18), 4 layers**: grounding prompt → **deterministic verbatim-quote verifier** (every quoted span must be a substring of retrieved context — catches fabrication at zero LLM cost) → score-threshold abstention → LLM-judge on a sample. Finding: the verifier must normalize whitespace *and* punctuation spacing (LLMs silently tidy quotes), else faithfulness is under-reported.
- **Query handling**: a light gate routes IN_SCOPE / OUT_OF_SCOPE / AMBIGUOUS (out-of-scope declined gracefully); multi-turn follow-ups are rewritten into standalone queries (*condense-question*, bounded last-N + summary memory) — fixing **retrieval**, not just generation, while grounding is untouched.

Considered and rejected (named): Corrective RAG — adopt assess-and-correct routing, reject web-search fallback (external text ≠ the executed contract); Cache-Augmented Generation — rejected as primary (off-track; tail contracts ≈75k tokens; blurs citation), adopted as session prompt-cache; Self-RAG / Adaptive-RAG / GraphRAG — ill-fitting a no-fine-tune single-hop task; RAGAS — we have real gold labels, so we measure retrieval rather than LLM-judge it (kept as optional cross-check).

3 · Results & Error Analysis (held-out CUAD test)

axis	winner	key numbers (244-query stratified subset)
retrieval mode (Q17)	hybrid-RRF	dense R@5 .47 · BM25 .54 · hybrid .55 · +rerank .48 (<i>hurt</i>)

chunking (Q15)	recursive-512	R@10: 256 = .665 · 512 = .753 ; structure-aware tied, not better
embedding (Q16)	bge-small	bge-small R@10 .665 ≈ bge-base .656 (no lift)

End-to-end (balanced 300-pair sample; winner config): answer token-F1 **0.58 when it answers** (0.44 incl. abstentions) · abstention **precision 0.73 / recall 0.83** · strict verbatim-verifier faithfulness **0.53**. Error taxonomy (300): clean-answer 48 · clean-abstain 125 · unfaithful-quote 46 · wrongful-abstention 46 · wrongful-answer 25 · correct-context-wrong-answer 10. Honest findings:

- **Faithfulness 0.53 is a strict lower bound**, not “hallucinates half the time”: the gate flags an answer if *any* quote is non-verbatim, and verbose answers pair a correct primary quote (F1 up to 1.0) with paraphrased “key-point” quotes that fail. Verifying only the primary citation, or suppressing embellishment, would raise it; the strictness is deliberate — a hard safety gate.
- **Wrongful abstention (46) is not purely retrieval miss**: a large slice is the “Document Name” category, where the gold answer is the contract *title* — retrieved, but the model declines to treat “the contract is called X” as the answer. Part retrieval recall, part question interpretation.
- **Abstention recall 0.83**: 25 unanswerables were answered (over-explaining a *related* clause); the deterministic verifier flags most as UNGROUNDED — a partial safety net when generation over-reaches.

Caveat: the retrieval ablation ran on a seeded 20-contract subset (CPU tractability; the relative ranking is what the ablation decides); the e2e eval used the full balanced 300-pair sample.

4 · Production & Limitations

Production (1,000 concurrent queries): bottlenecks are (a) embedding CPU → GPU batch server; (b) LLM rate limits → response caching + request queue; (c) FAISS flat → HNSW/pgvector. **Measured per-query latency** (logged for every query): retrieval ~0.2s CPU · generation ~3s (Haiku 4.5, ~1.2k in / ~0.2k out tokens) · gate ~2.6s (parallelizable) → ~6s total; Bedrock prompt-caching amortizes per-contract context across a session. **Freshness (demonstrated)**: adding a contract embeds only that contract — 0.8s, existing vectors untouched, immediately queryable; the real re-embed trigger is an embedding-model version change, not corpus growth.

Limitations: (1) tables/exhibits — text-only chunking loses tabular layout; (2) multi-span answers (mean 2.1, max 19 spans/question) stress single-chunk retrieval; (3) the single biggest lever is **retrieval recall** — a legal-domain reranker or fine-tuned embedder is the clear next step and would directly shrink the wrongful-abstention bucket.